Synopsis

Disease Prediction using Machine Learning

Project ID : 1

Problem ID : PY-GT-48

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1. Problem Description:

With the increasing availability of healthcare data, predicting disease risk at an early stage has become a crucial task in modern healthcare systems. Early detection helps in timely treatment, reduces medical costs, and improves patient outcomes.

In this project, an end-to-end machine learning classification system is developed to predict whether a person is at high risk of disease based on clinical and lifestyle parameters. The dataset includes features such as age, gender, blood pressure, smoking habits, and other relevant medical indicators.

The system performs data preprocessing, exploratory data analysis (EDA), feature engineering, feature selection, and model training to build an accurate prediction model. The final goal is to classify individuals into:

- High Disease Risk
- Low/Medium Disease Risk

This system can assist healthcareprofessionalsinmakingbetter decisions and promoting preventive care.

1.1 Dataset Description

The dataset used in this project is a healthcare dataset designed to predict disease risk based on various clinical and lifestyle parameters. It contains multiple features that represent an individual's health condition, habits, and physiological measurements.Each row in the dataset represents a single patient record, and each column represents a specific feature.

Features in the Dataset

- **Age (years):**
  Represents the age of the individual in years.
- **Gender:**
  Indicates the gender of the individual (Male, Female, Others).
- **BMI (kg/m²):**
  Body Mass Index, used to assess whether a person has a healthy body weight.
- **Blood Pressure (mmHg):**
  Recorded in systolic/diastolic format (e.g., 120/80), indicating cardiovascular health.
- **Cholesterol (mg/dL):**
  Measures cholesterol level in the blood.
- **Glucose (mg/dL):**
  Indicates blood sugar level.
- **Heart Rate (bpm):**
  Number of heartbeats per minute.
- **Smoking:**
  Indicates whether the individual smokes (Yes/No).
- **Alcohol Intake (units/week):**
  Weekly alcohol consumption.
- **Physical Activity (hours/week):**
  Amount of physical exercise performed per week.
- **Family History:**
  Indicates whether there is a family history of disease (Yes/No/Unknown).

Target Variable

- **Disease_Risk:**
  Represents the risk level of disease:
  - Low
  - Medium
  - High

For model training, this variable is converted into binary form:

- **0 → Low / Medium Risk**
- **1 → High Risk**

Dataset Characteristics

- Contains both **numerical and categorical features**
- Includes **realistic variations**such as:
  - Missing values
  - Inconsistent formats (e.g., "Yes", "yes", "Y")
- Blood pressure is stored in **combined format**, later split into:
  - Systolic
  - Diastolic

Purpose of Dataset

The dataset is used to:

- Analyze relationships between health factors and disease risk
- Train machine learning models for classification
- Predict whether a person is at high risk of disease

Sample dataset-

Age (years)	Gender	BMI (kg/m^2)	Blood_Pressure (mmHg)	Cholesterol (mg/dL)	Glucose (mg/dL)	Heart_Rate (bpm)	Smoking	Alcohol_Intake (units/week)	Physical_Activity (hours/week)	Family_History	Disease_Ris
34	male	27.9	94/119	277	104	111		4	5.8	no	Low
72	Female	25.7	163/109	264	122	81		4	6.4	Yes	High
29	Others		161/108	248	103	90	no	1	3.3	unknown	High
18	female	27.9	115/112	191	127	64	N	6	9.1	N	
34	male	25.5	169/117	152		79	Yes	6	8.4	N	Medium
19	Others	19	106-68	160	163	56	No	1	6.7	Yes	High
27	F	33.8	139/83	174	170	113	N	4	0	yes	Medium
18	Others	36.3	94-63			58		8	5.5	N	Low
68	Female	32.4	164/99	258	175	71	No	7	1.3	N	High
39	Others	18.2	166/62		149	90	N	4	2.7	Yes	High
30		30.8	127/101	246	144	78	no	3	9.8	Y	Medium
32		20.5	159-75	244	124	118	Y	0	3.5	Y	Low
20	Others	21.3	134-81	236	107	112		9	8.3	Y	Low
18	M	32.6	127/92	193	137	96	No	11	5.3	no	Low
22	male	29.5	107/72	248	89	105	yes	11	9.5	yes	Medium
22	Others	24.6	100/101	222	191	91	No	6	8.9	No	Low
35		29.8	151/119	212	73	112	Y	2	9.7	No	Low
38	M	33.8	106/118	180	141	70	Yes	2	2.5	yes	Low
19	Male	26.8	147/112	228	170		Yes	9	9.3	yes	Low
75	Others	36.7	162/116	245	88	113	N	5	1	Yes	Medium
20	male	16.5	151/83	254	86	113	Y	5	1.4	Y	Medium
22	Male	30.3	168/119	311	160	97	yes	4	3	no	High
22	Female	17.1	139/66	174	156	55	No	10	8.3	no	High
21	M		129/61	273	190	64	no	4	2.7	yes	High
68	M	25.1	96-76	244	122	118	no	9	2.1	Y	
41	F	23.5	109/119	172	181	88	N	5	1.7	Yes	Medium
	male	28.1	143/91	293	193	68	Yes	2	3.5		Medium
45	Others	26.3	154/84	223	129	79	Yes	2	0.7	no	High
21	Others	25.7	175/86	243	83	81	Yes	7	5.2	Y	Low

The dataset used for this project is a synthetically generated Wi-Fi network dataset, designed to closely mimic real-world network conditions. It includes variations, noise, missing values, and outliers to simulate practical scenarios and enable realistic data analysis.

1.2. Flow Diagram:

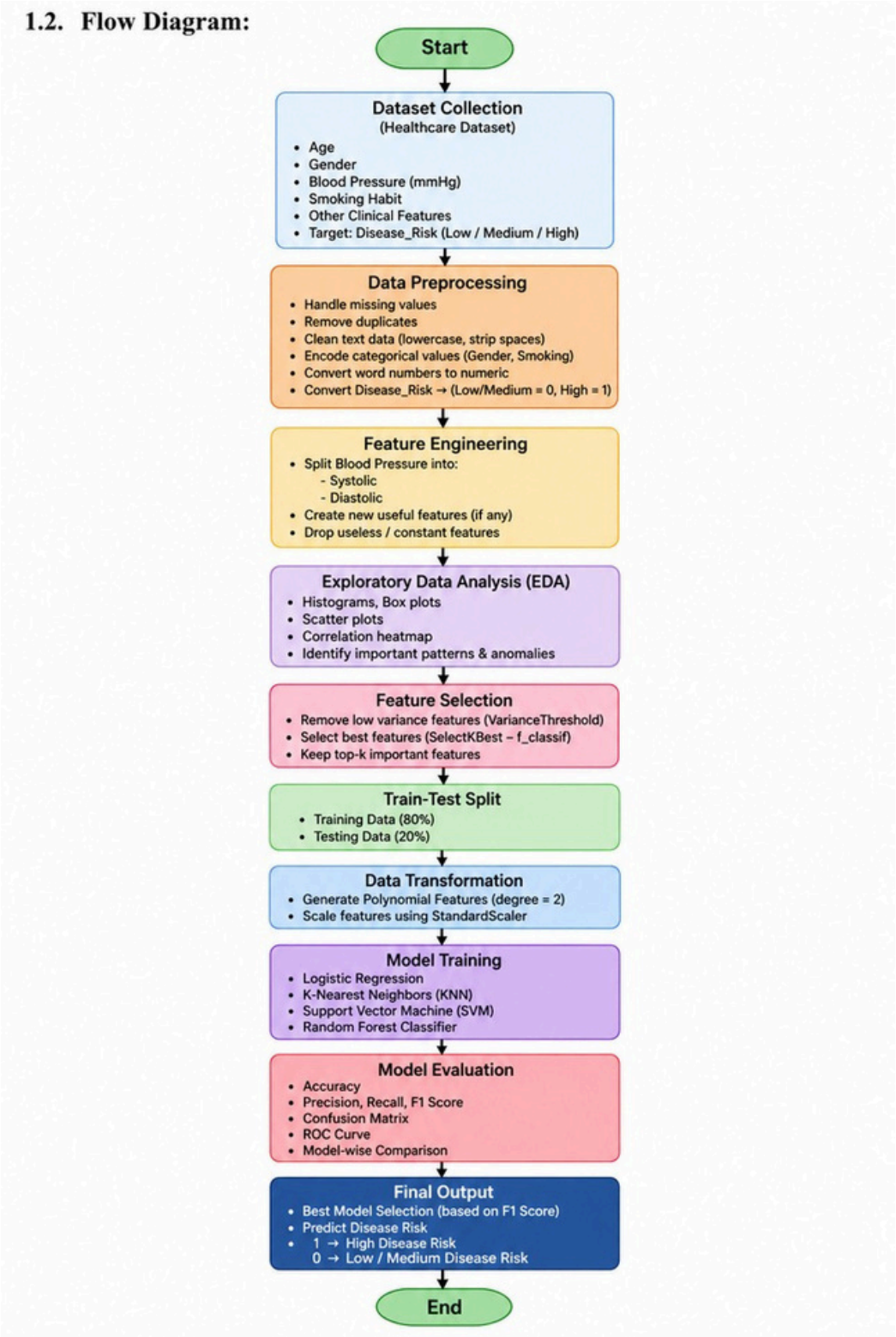


Fig. 1: Flow diagram of the proposed system.

2. Modules for Project Implementation:

2.1 Modules Used:

Pandas

Pandas is used for data manipulation and analysis, enabling efficient handling of structured data. It helps in cleaning, organizing, and summarizing the Wi-Fi dataset. For example, it can be used to compute average latency or filter rows with high packet loss.

NumPy

NumPy supports numerical computations and array operations. It is used for mathematical transformations and scaling of features such as latency and bandwidth usage before model training.

Matplotlib

Matplotlib is used for basic data visualization such as histograms and line plots to understand data distribution and trends in network usage.

Seaborn

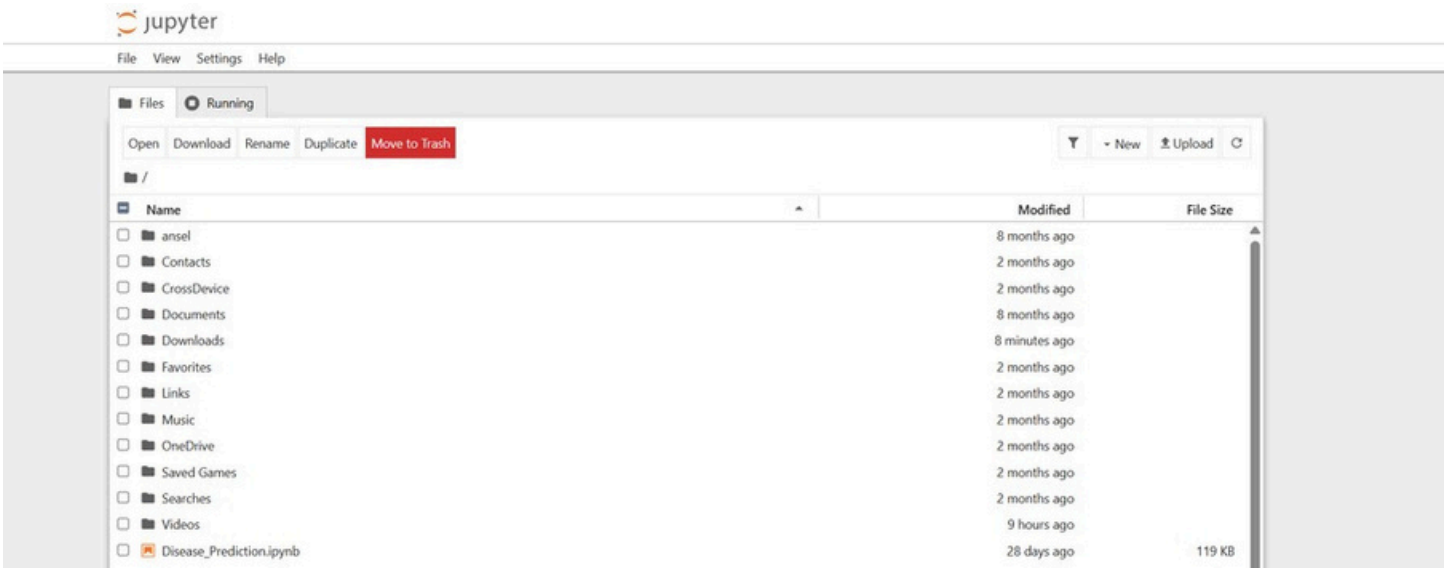
Seaborn provides advanced visualizations like heatmaps and box plots. It is useful for correlation analysis between network features and identifying congestion patterns.

Scikit-learn

Scikit-learn is used to build and evaluate machine learning models including Logistic Regression, Random Forest, and SVM. It also helps in preprocessing and splitting the dataset.

2.2. Platform Used:

Jupyter Notebook is used as the primary platform for implementing the project. It provides an interactive environment for writing and executing Python code, performing data preprocessing, and conducting Exploratory Data Analysis (EDA). It also supports visualization libraries like matplotlib and seaborn, making it suitable for analyzing and presenting Wi-Fi network data effectively.

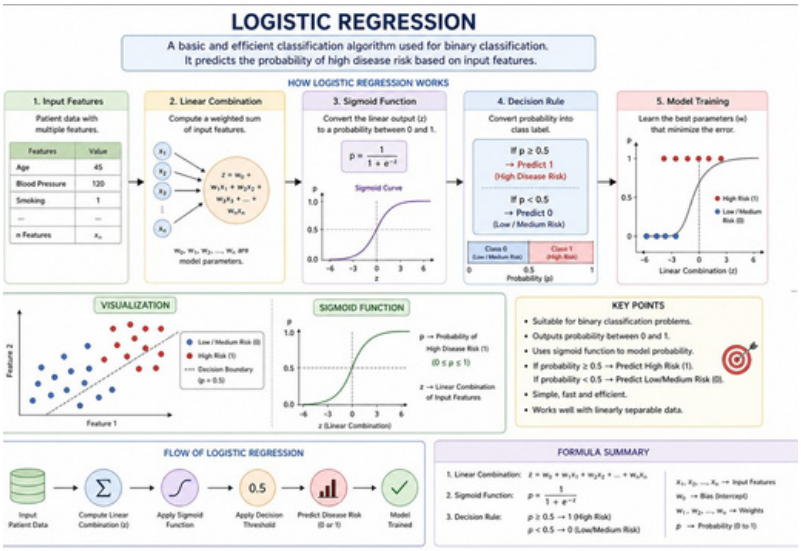


```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# ---------------- LOAD DATA ----------------
df = pd.read_csv("Improved_DataSet.csv")

print("Columns:", df.columns)
print(df.head())

# ---------------- TARGET CLEANING FIRST ----------------
if "Disease_Risk" not in df.columns:
    raise ValueError("Target column 'Disease_Risk' not found")

# Convert target to numeric
df["Disease_Risk"] = df["Disease_Risk"].astype(str).str.strip().str.lower().map({
    "low": 0,
    "medium": 0,
    "high": 1
})

# Save target separately
y = df["Disease_Risk"]

# ---------------- FEATURE CLEANING ----------------
```

3. Machine Learning Models Used

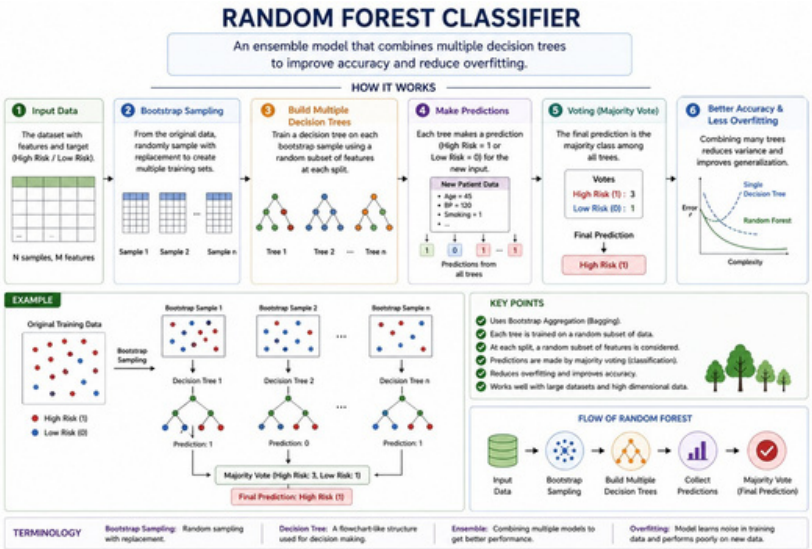
Logistic Regression

A statistical classification algorithm that predicts the probability of Wi-Fi congestion using a sigmoid function. It is simple, fast, and effective for baseline performance in binary classification problems.



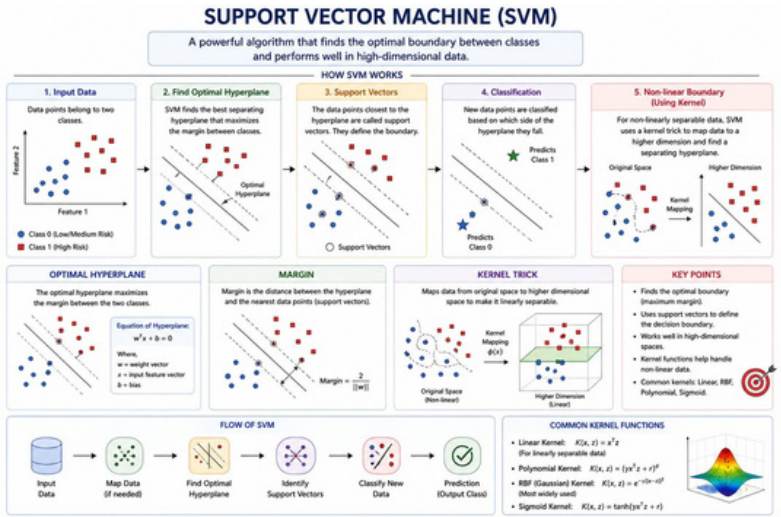
Random Forest Classifier

An ensemble learning technique that builds multiple decision trees and combines their outputs for better accuracy. It handles complex relationships between features like latency, packet loss, and bandwidth, and reduces overfitting.



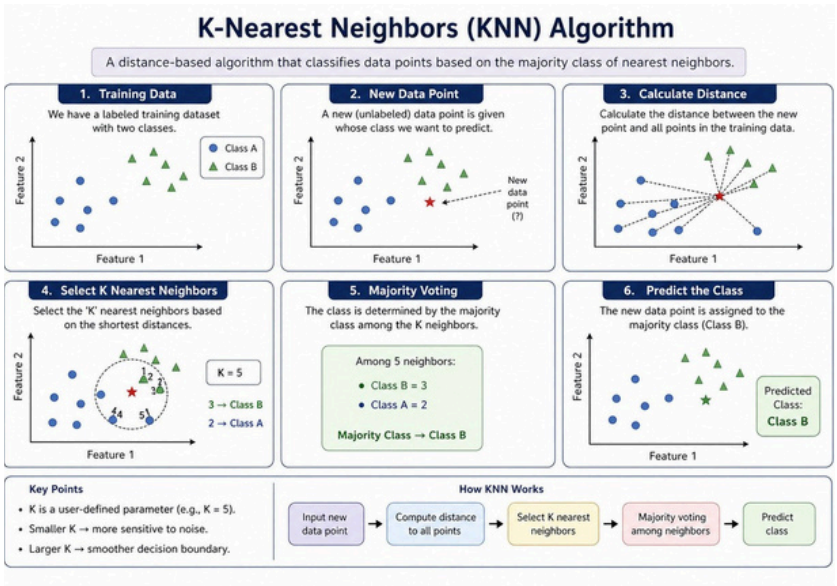
Support Vector Machine (SVM)

A powerful algorithm that finds the optimal hyperplane to separate congested and non-congested classes. It performs well in high-dimensional data and can model non-linear boundaries using kernel functions.



K-Nearest Neighbors (KNN)

A distance-based algorithm that classifies data points based on the majority class of nearest neighbors.



4. Evaluation Metrics Used

Accuracy

Accuracy measures the percentage of correctly predicted instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Confusion Matrix

A confusion matrix shows:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Importance:

- Helps analyze model performance
- Identifies misclassification

Heatmap (Confusion Matrix Visualization)

A heatmap visually represents the confusion matrix using colors.

Advantages:

- Easy interpretation
- Detects errors quickly

5. Results And Visualizations

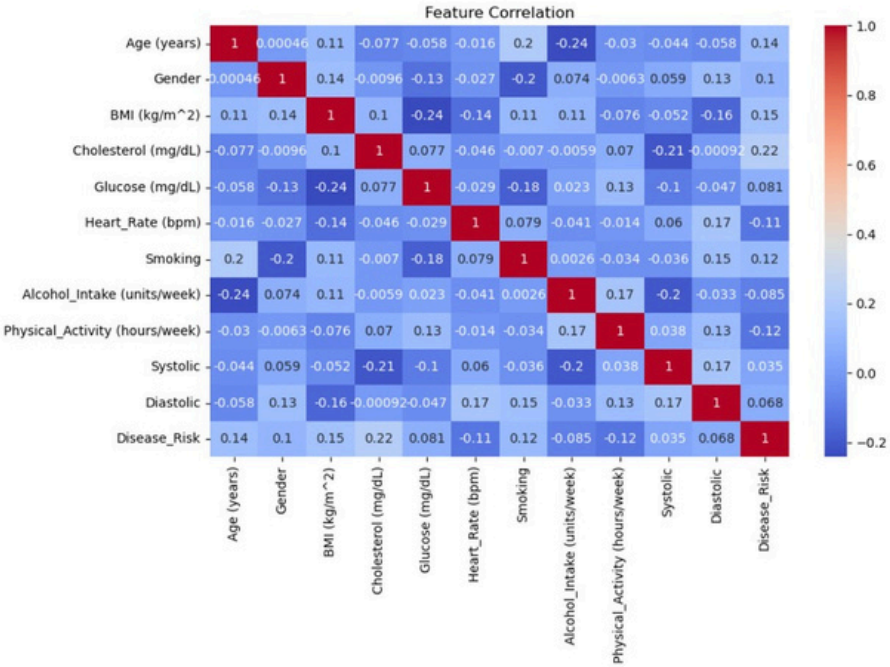


Fig 2: Figure Showing Heatmap of Feature Correlation

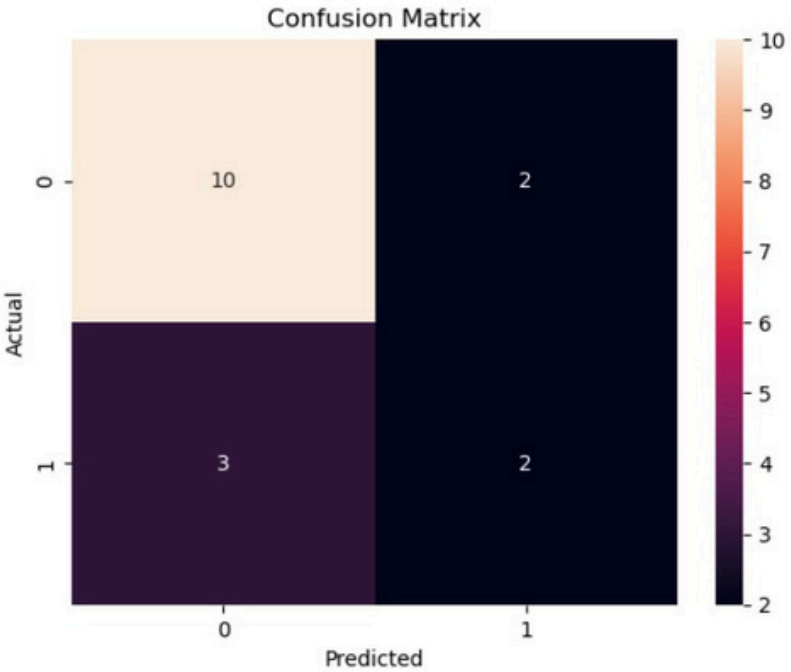


Fig 3: Figure Showing Confusion Matrix

6. Conclusion and Future Scope

Conclusion

This project successfully developed a machine learning model for disease risk prediction.

- Data preprocessing improved data quality
- Feature engineering enhanced performance
- Multiple models were trained and compared
- Best model selected based on evaluation metrics

The system demonstrates the importance of machine learning in healthcare prediction systems.

Future Scope

- Use real-world medical datasets
- Increase dataset size for better accuracy
- Apply deep learning models
- Deploy as a web or mobile application
- Integrate with hospital systems

7. References

Books

- **Let Us Python – Yashavant Kanetkar**

Online Resources

1. **Scikit-learn Official Documentation (ML Models & Preprocessing)**

https://scikit-learn.org/stable/

Covers data preprocessing, feature selection methods (PCA), and classification algorithms like Logistic Regression, SVM, KNN, etc.

1. **Kaggle Learn (Practical ML & EDA Tutorials)**

https://www.kaggle.com/learn

Offers hands-on tutorials on data preprocessing, EDA, feature engineering, and machine learning workflows.

1. **Official Seaborn Heatmap Documentation**

https://seaborn.pydata.org/generated/seaborn.heatmap.html

Explains how to use heatmaps in seaborn library

1. **Chat GPT**

https://chatgpt.com/